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AI Requirements Framework

for Safety-Critical Systems

Part 2: Classification Determination, Evidence Templates and Revision Record

Safety Critical Labs
AI CERTIFICATION AUTHORITY

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Table of Contents

Section 1: Status and Use 2
Section B: AI Classification Determination 3
B.1 AI PRESENCE DETERMINATION 3
B.2 SAFETY-CRITICAL AI DETERMINATION 4
B.3 MISSION-CRITICAL AI DETERMINATION 4
B.4 OPERATIONAL DESIGN DOMAIN DECLARATION 5
B.5 PRIVACY APPLICABILITY DETERMINATION 5
B.6 REQUIREMENT APPLICABILITY 5
Section C: Verification Evidence Templates 6
C.1 AI APPLICABILITY DETERMINATION RECORD 6
C.2 TEST REPORT TEMPLATE 7
C.3 INSPECTION RECORD TEMPLATE 7
C.4 ODD DECLARATION RECORD TEMPLATE 8
C.5 PRIVACY IMPACT ASSESSMENT TEMPLATE 9
C.6 DATA SUBJECT RIGHTS REQUEST LOG TEMPLATE 9
C.7 SUFFICIENCY STATEMENT TEMPLATE 10
C.8 DECLARED EVALUATION POPULATION RECORD 11
Annex: Revision Record 12

Section 1: Status and Use

This document is Part 2 of the SCL AI Requirements Framework for Safety-Critical Systems. Part 1 (SCL-ARF-001) defines the framework: the requirements, the verification statements, the verification matrix and the conformity basis. This Part carries what supports applying it and nothing that defines a requirement: the AI classification determination (B.1 to B.6), the verification evidence templates (C.1 to C.8), and the revision record. It is issued with Part 1 at the same version number and in the same deposit, so that a version DOI resolves to both parts, and it is cited from Part 1 Table 1-9.

Normative status. B.1 to B.6 and the required content of the records C.1 to C.8 are normative to the extent Part 1 states where it cites them; they form part of the

conformity basis of Part 1 Section 1.6. The revision record is informative. Where this document and Part 1 differ, Part 1 governs.

Identifiers. The section identifiers B.1 to B.6 and C.1 to C.8 are those the same material carried as appendices of Part 1 in versions 3.x, retained so that references in evidence packages, scheme documents and the website resolve unchanged. Requirement identifiers (AI-x.y), section numbers, table numbers and the glossary cited in this document are those of Part 1 at the same version.

Section B: AI Classification Determination

This checklist supports determination of AI system classification per Part 1 Section 1.2.

B.1 AI PRESENCE DETERMINATION

Before answering, declare the determination boundary per Part 1 Section 1.2.1 (the bounded collection of hardware and software items under determination and its declared interfaces) and record it in C.1. The declaration shall name the encompassing system and enumerate every candidate learned component within it; every candidate learned component shall lie within a declared determination boundary, and a negative determination discharges only the boundary for which it was made. The determination is evidence-cited: each answer shall record the artifact or artifacts inspected (system or software requirements, design descriptions, interface control documents, model provenance records, data flow documentation). Answer question 1 first. Questions 2 and 3 record supporting indicators and do not establish AI presence on their own.

1. (Anchor) Within the declared boundary, is any system behavior specified by an artifact induced from data rather than by an explicitly written specification against which correctness can be fully verified by established methods? For behavior implemented through data-fitted parameters, apply the deletion test (Part 1 Section 1.2.1): delete every value derived from data, or designed to be derived from data during operation, and inspect what remains; a surviving behavioral specification (written structure bounding behavior, values re-derivable by a documented derivation constituting a correctness argument, or generated at runtime by an explicitly specified procedure whose established analysis bounds the correctness of the resulting behavior) means the anchor criterion is not met, and no surviving specification means it is met. Parameter determination or improvement by machine learning, including retraining and continuous learning during operation, meets the anchor criterion regardless of when the optimization

runs. Cite the evidence inspected. If 'No' → not AI for framework purposes; standard software assurance applies; record the negative determination and its evidence in C.1 (see Part 1 Table 1-2b for exclusion examples). If 'Yes' → AI is present; answer questions 2 and 3, then proceed to B.2.

2. (Supporting indicator) Are system outputs probabilistic or non-deterministic? If 'Yes' → note verification emphasis on confidence indication and calibration (AI-8.3). Cite the evidence inspected.

3. (Supporting indicator) Can system behavior change or degrade over time without code modification? If 'Yes' → the system shall be classified no lower than Mission-Critical per Part 1 Section 1.2.2. Cite the evidence inspected.

Evaluation: Question 1 alone determines AI presence. If question 1 is 'Yes' → AI is present; proceed to B.2. If question 1 is 'No' → this framework does not apply regardless of the answers to questions 2 and 3; standard software assurance applies. A determination is complete only when the determination boundary is declared and each answer cites the evidence inspected; the record is maintained in C.1.

B.2 SAFETY-CRITICAL AI DETERMINATION

1. Do AI outputs directly affect human safety or system survivability?

2. Could AI malfunction result in loss of life or critical equipment?

3. Is the AI in the control path for safety-critical functions?

Evaluation: If any of questions 1 through 3 above is 'Yes' → Safety-Critical AI. If all are 'No' → proceed to B.3.

Each answer shall be supported by the four classification questions of Part 1 Section 1.2.2 (the consequence of a wrong output; whether a human can detect and correct it before harm; control path, advisory path or neither; latency to consequence), recorded with evidence in the Applicability Determination Record. The Lead Assessor's determination governs where it differs from the applicant's.

B.3 MISSION-CRITICAL AI DETERMINATION

1. Do AI outputs affect operational success (but not human safety)?

2. Could AI malfunction result in loss of operational objectives?

3. Is the AI in the decision path for mission-critical functions?

Evaluation: If any of questions 1 through 3 above is 'Yes' → Mission-Critical AI. If all are 'No' → Operational Support AI.

Each answer shall be supported by the four classification questions of Part 1 Section 1.2.2 (the consequence of a wrong output; whether a human can detect and correct it before harm; control path, advisory path or neither; latency to consequence), recorded with evidence in the Applicability Determination Record. The Lead Assessor's determination governs where it differs from the applicant's.

B.4 OPERATIONAL DESIGN DOMAIN DECLARATION

Following classification determination, declare the ODD per AI-1.0. Confirm the following:

- Operational Claim summary statement drafted
- All eight ODD attribute categories addressed (Operational Environment, System Inputs, User Population, Subject Population, Operational Phases, Performance Envelope, Regulatory and Operational Authorization, Exclusions)
- Exclusions enumerated across all five sub-categories (Environments, Use Cases, Operator Conditions, Subject Conditions, System States)
- Declaration approved at the authority level specified for the system's classification per Part 1 Section 1.2.3 Part 1 Table 1-4

B.5 PRIVACY APPLICABILITY DETERMINATION

- Does the system process personal data as defined under any applicable privacy framework (GDPR, HIPAA, CCPA/CPRA, PIPEDA, or analogous)?
- Does the system process special categories of personal data (e.g., GDPR Article 9, HIPAA PHI, CCPA sensitive personal information)?
- Does the system transfer personal data across jurisdictional boundaries?

Evaluation: If any answer is 'Yes' → AI-10 applies in full; conduct PIA/DPIA per AI-10.6. If all answers are 'No' → AI-10 sub-requirements may be documented as Not Applicable with rationale.

B.6 REQUIREMENT APPLICABILITY

Classification	Applicable Requirements
Safety-Critical AI	All AI-1 through AI-10 (Core Requirements), AI-5 where the system produces generative or free-form output and AI-10 where personal data is processed; AI-14 (Runtime Assurance) mandatory; plus AI-11 if multi-model, AI-12 if neural network (AI-12.2 and AI-12.4 also apply to other statistical machine learning models), AI-13 if continuous

Classification	Applicable Requirements
	learning (shadow mode only at this tier), and AI-15 where an information classification regime applies
Mission-Critical AI	AI-1 through AI-4, AI-6, AI-8, AI-9 (Core Requirements); AI-5 where generative or free-form output; AI-10 where personal data is processed; AI-7 tailored, with the threat-model reason stated in the tailoring record; AI-14 available by tailoring-in; plus AI-11, AI-12, AI-13 and AI-15 conditionally per system design and regime
Operational Support AI	AI-1 through AI-4 and AI-6 (Core Requirements) minimum; AI-5 where generative or free-form output; AI-10 where personal data is processed; AI-7, AI-8, AI-9 as tailored; plus AI-11, AI-12, AI-13 and AI-15 conditionally per system design and regime

Section C: Verification Evidence Templates

C.1 AI APPLICABILITY DETERMINATION RECORD

Field	Entry
Project Name	
System/Subsystem	
Determination Boundary (B.1)	
Document ID	
Date	
Prepared By	
AI Presence (B.1 Result)	☐ AI Present ☐ No AI
Evidence Cited (B.1)	
AI Classification (B.2/B.3)	☐ Safety-Critical ☐ Mission-Critical ☐ Operational Support
Applicable Requirements	☐ AI-1 ☐ AI-2 ☐ AI-3 ☐ AI-4 ☐ AI-5 ☐ AI-6 ☐ AI-7 ☐ AI-8 ☐ AI-9 ☐ AI-10 ☐ AI-11 (conditional) ☐ AI-12 (conditional) ☐ AI-13 (conditional)
Foundational Cybersecurity Baseline	☐ Attestation ☐ Existing conformity evidence (identify)

Field	Entry
(Part 1 Section 1.6)	
Tailoring	☐ None ☐ See attached rationale
Approved By	
Approval Date	

C.2 TEST REPORT TEMPLATE

Field	Entry
Test Report ID	
Requirement ID(s)	
Test Title	
Test Date	
Test Environment	
Test Conductor	
Test Objective	
Pass/Fail Criteria	
Declared Set and Derivation Basis (AI-7.1, AI-7.2, AI-10.4)	
Test Results	
Pass/Fail Determination	☐ Pass ☐ Fail
Anomalies/ Observations	
Approved By	

C.3 INSPECTION RECORD TEMPLATE

Field	Entry
Inspection Record ID	
Requirement ID(s)	
Artifact Inspected	
Inspection Date	
Inspector	

Field	Entry
Inspection Criteria	
Findings	
Compliance Determination	☐ Compliant ☐ Non-Compliant ☐ Partially Compliant
Corrective Action (if any)	

C.4 ODD DECLARATION RECORD TEMPLATE

Field	Entry
Project Name	
System/Subsystem	
Document ID	
Classification	☐ Safety-Critical ☐ Mission-Critical ☐ Operational Support
Operational Claim	*This AI system is designed and validated to [function] for [subject population] under [operational conditions], operated by [user role] within [regulatory regime].*
Operational Environment	
System Inputs (including degradation behaviors)	
User Population	
Subject Population	
Operational Phases	
Performance Envelope	
Regulatory and Operational Authorization (including ODD evolvability)	
Excluded Environments	
Excluded Use Cases	
Excluded Operator Conditions	
Excluded Subject Conditions	
Excluded System States	
Approved By	

Field	Entry
Approval Date	
Review Cadence	

C.5 PRIVACY IMPACT ASSESSMENT TEMPLATE

Field	Entry
Project Name	
System/Subsystem	
PIA ID	
Applicable Frameworks	☐ GDPR ☐ HIPAA ☐ CCPA/CPRA ☐ Other: _____
Personal Data Categories Processed	
Special Category Data	☐ Yes ☐ No (if Yes, categories: _____)
Lawful Basis (per framework)	
Data Flow Map Reference	
Cross-Border Transfers	☐ None ☐ See attached transfer documentation
Privacy-Enhancing Techniques Applied	
Re-Identification Risk Assessment	
Model Inversion and Membership Inference Risk	
Data Subject Rights Implementation Summary	
Identified Privacy Risks and Mitigations	
Approving Privacy Authority	
Approval Date	
Next Review Date	

C.6 DATA SUBJECT RIGHTS REQUEST LOG TEMPLATE

Field	Entry
Request ID	

Field	Entry
Receipt Date	
Request Type	☐ Access ☐ Rectification ☐ Erasure ☐ Restriction ☐ Portability ☐ Objection ☐ Automated Decision Explanation
Applicable Framework	☐ GDPR ☐ HIPAA ☐ CCPA/CPRA ☐ Other: _____
Requestor Verification Method	
Applicable Response Timeline	
Processing Steps	
Model Retraining Assessment (erasure only)	
Disposition	☐ Granted ☐ Partially Granted ☐ Denied (with rationale)
Response Date	
Response Delivery Method	
Approved By	

C.7 SUFFICIENCY STATEMENT TEMPLATE

Field	Entry
Requirement set	Identifier and title (one statement per applicable set)
Parent outcome claimed	The [R.AI-x] will statement, restated for the declared Operational Design Domain
Evidence, deterministic class	Sub-requirements closed, with artifact references
Evidence, statistical class	Sub-requirements supported, with interval, confidence and declared population for each
Evidence, bounded class	Sub-requirements demonstrated, with the budget for each
Evidence, surveillance class	Obligations in service, with cadence and the indicator they monitor

Field	Entry
Residuals	Insufficient-evidence outcomes, bounded budgets and outstanding surveillance obligations, each with the tier rule of Part 1 Section 1.6 applied
Compensation for each residual	The requirement, architectural element (AI-14 at Safety-Critical) or operational restriction that compensates, by identifier
Statement	One paragraph: why the evidence listed supports the parent outcome for the declared ODD, and what it does not establish
Prepared by / date	Name, role, date; approval authority per Part 1 Section 1.2.2

C.8 DECLARED EVALUATION POPULATION RECORD

Field	Entry
Population or set identifier	Unique identifier used in the verification evidence
Used by	The [V.AI-x.y] statements and Part 1 Section 4.6 indicators that measure a rate on it
Definition	Membership rule, source, and the ODD attributes (Part 1 Table 2.1.0-A) it is drawn from
Strata	Required strata per Part 1 Section 1.7 (far / near boundary; injected / model-produced; Subject Population strata; grid cells; attack set and budget) with the governing stratum named
Boundary validation	For out-of-distribution strata, how membership outside the declared envelope was established before use (AI-6.1)
Size and power	Sample size per stratum; the rate it can support at the stated confidence (Part 1 Section 1.5)
Derivation basis	Why this population represents the declared ODD for the rates measured on it
Window and sample floor	For monitored indicators, the monitoring window and the floor below which the indicator is unavailable (Part 1 Section 4.6.1)
Certified configuration reference	Entry in the certified configuration (Part 1 Section 1.6) and in the Certification Determination Document

Field	Entry
Prepared by / date	Name, role, date

Annex: Revision Record

The full description of every version of the standard, as carried in the revision history of Part 1 through version 3.7 and continued here. Part 1 keeps a one-line summary per version; from version 3.4 each version also has a change log. The Author column names Safety Critical Labs for every version: the framework is the organization's, whoever drafted a given version.

Version	Date	Author	Change Class	Description
1.0	February 2025	Safety Critical Labs	Initial release	Initial framework development
2.0	October 2025	Safety Critical Labs	Normative Addition	Added AI-4.6 Operational Validation Without Ground Truth, AI-9.6 Operational Role Verification, Appendix D Risk Score Methodology
2.1	January 2026	Safety Critical Labs	Normative Addition	Added AI-1.0 Operational Design Domain with Operational Claim and eight attribute categories; AI-1 renamed to Operational and Data Foundations; added Section 1.2.3 ODD procedural gate and Table 1-4 ODD Declaration Rigor; added AI-10 Privacy and Data Protection with seven sub-requirements; added Section 1.3.1 Domain Standards Citation Convention with anchor citations on all ten parent requirements and exemplar sub-requirement retrofits on AI-9.1, AI-8.3, AI-2.1; added cross-references for ODD, AI-10, and model integrity across Section 2; expanded Section 4.2 Verification Matrix to cover all

Version	Date	Author	Change Class	Description
				sub-requirements; added Appendix C templates for ODD Declaration Record, Privacy Impact Assessment, and Data Subject Rights Request Log; added Appendix A glossary terms for ODD, Operational Claim, Subject Population, and privacy terminology; added Appendix B.4 ODD Declaration step and B.5 Privacy Applicability Determination; removed Appendix E Source Traceability Matrix (traceability retained inline in Appendix A); renamed the Mission Support classification tier to Operational Support throughout the framework (tier definition, tailoring authority, applicability, and worked examples).
2.1.1	April 2026	Safety Critical Labs	Normative Addition	Added Gap 6 human factors expansion to AI-9: AI-9.7 Operator Qualification, AI-9.8 Workload Management, AI-9.9 Training Program Requirements, each with full requirement, verification, success criteria, and Applicable Domain Standards citations. Added Gap 10 public disclosure support as new AI-8.5 Public Disclosure Support (optional by classification) covering external model cards, EU AI Act Article 13 user information, EO 14110 and OMB M-24-10 federal AI inventory, and EU AI Act Article 71 public database registration. Added corresponding rows to the Section 4.2 Verification Matrix. Sub-requirement-level Applicable Domain Standards retrofit for the

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				approximately 53 remaining sub-requirements is scheduled as a Phase 2 pass within v2.1.1.
2.1.3	05/04/2026	Safety Critical Labs	Normative Addition	Added AI-4.7 Deployment Format Validation. Added Section 4.0 Overview, Section 4.4 Continuous and Periodic Verification, and Cadence column to Section 3.2 Verification Matrix. Editorial pass on Sections 1-3 (terminology consistency, lexicon alignment, tracked-change cleanup).
3.0	05/05/2026	Safety Critical Labs	Normative Modification	Major structural revision. Added Section 3 (Architecture and Paradigm Requirements) with AI-11 Multi-Model Systems, AI-12 Neural Networks, AI-13 Continuous Learning and Adaptation. Renumbered prior Section 3 (Verification) to Section 4 and prior Section 4 (Implementation Guidance) to Section 5 (Implementation Patterns). Lifted normative content from prior Section 4.2/4.3/4.4 into new Section 3 architecture/paradigm requirements with proper [R.AI-X.Y] / [V.AI-X.Y] scaffolding. Resolves prior lexicon contradictions in Section 4 by separating normative architecture/paradigm requirements from informative implementation patterns.
3.1	06/11/2026	Safety Critical Labs	Editorial	Editorial and formatting revision. Bolded Section 3 verification statements ([V.AI-11.x] through [V.AI-13.x]) for consistency with

Version	Date	Author	Change Class	Description
				Sections 1-2. Normalized sub-heading capitalization to title case. Removed residual editorial annotations. Replaced em and en dashes per house style. Corrected fused sentences, duplicated text in Table 1-3, and assorted typos. Normalized Rationale and Success Criteria formatting. Updated Section 4.2 matrix cadence for AI-4.1 and AI-6.4 to Continuous. Extended Appendix C.1 requirement checklist through AI-13. Added Section 3 sub-requirements (AI-11.1 through AI-13.8) to the Section 4.2 Verification Matrix and the conditional AI-11/AI-12/AI-13 applicability to Appendix B.6, aligning both with Table 1-3. Added 1.3.3 to the table of contents.
3.2	06/14/2026	Safety Critical Labs	Editorial	Reformatted Section 3 (AI-11, AI-12, AI-13) verification blocks to the four-part Section 2 pattern (rationale, verification statement, method elaboration, success criteria) and aligned AI-4.7 phrasing. Verification format now uniform across AI-1 through AI-13.
3.3	06/14/2026	Safety Critical Labs	Editorial	Corrected the circular Test definition in the Section 4.1 verification-methods table; refined the Analysis and Demonstration definitions to sharpen the quantitative-Test versus qualitative-Demonstration distinction; added a Test entry to the Appendix A glossary; updated the IEEE 1012 citation to the 2024

Version	Date	Author	Change Class	Description
				edition; corrected a stale section cross-reference in the normative-references table.
3.4	07/08/2026	Safety Critical Labs	Normative Addition; Normative Modification	Redesigned the Section 1.2.1 applicability gate: Learned behavior is the anchor criterion, Probabilistic outputs and Potential drift are supporting indicators, Decision impact relocated to classification in Section 1.2.2; added Table 1-2b negative criteria with worked examples; rewrote Appendix B.1 as a decision sequence and gave B.1 through B.3 self-contained question numbering. Added AI-1.9 AI Hazard Analysis Integration and AI-1.10 Retrieval Corpus Controls with verifications and matrix rows. Added Section 4.5 Verification Independence scaled by classification. Added classification severity anchoring and approval authority to Section 1.2.2. Added threshold documentation and adequacy language to Section 1.5. Extended AI-7.2 to prompt injection for generative and retrieval-augmented systems. Extended AI-12.2 and AI-12.4 to non-neural-network statistical machine learning models. Reworded AI-2.1 and AI-2.2 from bias-free to bias-bounded and bias-managed; corrected AI-2.3 scope to include Safety-Critical decisions; added fallback response paths to AI-2.11 and AI-5.3. Assigned the Demonstration method to AI-8.2 and AI-9.3 and

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				aligned logging and calibration verification cadences. Corrected stale section references and miswired cross-references in Sections 1.4, 2, 4.0, 4.4, 5.3.2, AI-4.2, AI-8.5, AI-9.7, and AI-9.9. Added the Table 1-7 Retirement phase, the UL 4600 reference, Not Applicable provisions for AI-1.5 through AI-1.8, Table 1-4 rigor scaling in AI-1.0, and an input space coverage exemplar replacing neuron activation coverage in Section 5.1.1.
3.5	07/29/2026	Safety Critical Labs	Normative Modification	Replaced the undefined term “host standard” in AI-1.9 (requirement, rationale, and verification) with the project’s system safety process; added the System Safety Process definition to the Appendix A glossary, distinguishing the system-level hazard analysis process from the software standard this framework supplements; amended Section 1.2.5 to identify AI-1.9 as the framework’s single interface requirement to the system safety process. Findings from the 2026-07-28 correspondence review, verified against the v3.4 text.
3.6	08/08/2026	Safety Critical Labs	Normative Addition	Made the Section 1.2.1 applicability determination verifiable. Promoted the key-existence principle into the anchor criterion: applicability turns on whether correctness can be fully verified against explicitly written specification by established methods, with the Table 1-2 mechanism list illustrative rather

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				than load-bearing. Defined the determination boundary as the unit of determination, adapted from the EASA AI Concept Paper Issue 2 AI/ML constituent concept, with an explicit propagation rule: the framework attaches to learned components and to artifacts whose correctness criterion is fidelity to a learned component, and ends at declared interfaces. Adopted the deletion test for data-fitted parameters as a normative determination step in Appendix B.1 with a corresponding Table 1-2b exclusion row, incorporating runtime-generated-parameter and derivation-as-correctness-argument provisos validated by a seven-case battery (GRAM, wind tunnel aerodynamic databases, gain scheduling, PID autotune, adaptive Kalman filtering with online covariance estimation, table-driven FADEC, learned-then-frozen gain tables). Added a Monte Carlo simulation and dispersion analysis row to Table 1-2b. Reworked Appendix B.1 as an evidence-cited determination with declared boundary; added Determination Boundary and Evidence Cited fields to Appendix C.1. Added the Determination Boundary glossary entry, the EASA AI Concept Paper Issue 2 reference to Table 1-10, and ARP6983/ED-324 to the Section 1.3.3 unverified citations list. Same-day review pass: added the

Version	Date	Author	Change Class	Description
				determination boundary coverage condition; added Learned Artifact and Learned Component glossary definitions grounded in ISO/IEC 22989 process terms, clause-verified against the source standard; made the deletion test timing-neutral to cover learning during operation; conditioned the runtime parameter proviso on correctness-bounding analysis with an explicit machine learning and continuous learning carve-out; added an adaptive-variant pointer to the classical state estimators row; corrected the Appendix A Parameter definition to internal variable per 3.3.8 and re-attributed the Artificial Intelligence entry as an adaptation of 3.1.4.
3.7	09/15/2026	Safety Critical Labs	Normative Addition; Normative Modification (Appendix D.5)	Added Section 1.6 Conformity Basis, stating that the certification determination rests on the framework's requirement statements verified per their paired verification statements and on nothing outside the document, that domain standards crosswalks carry no conformity burden, that the foundational cybersecurity posture is a declared prerequisite carried on the certificate rather than assessed scope, and defining the certified configuration. Partitioned the references: Table 1-9 now lists only the normative references with the extent of each invocation (ISO/IEC 22989:2022, ISO/IEC

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				5338:2023, IEEE 1012-2024) and Table 1-10 the informative references; renamed every "Applicable Domain Standards" subsection to "Related Domain Standards" and restated the citation convention in Section 1.3.1, naming Safety Critical Labs as the party that performs the crosswalk during assessment. Replaced "known attack patterns" in V.AI-7.1 and V.AI-7.2 with a project-declared attack pattern set documented with its derivation basis, adjudicated during assessment, and recorded as an element of the certified configuration; applied the same construction to the indirect prompt injection patterns in V.AI-1.10 and the cryptographic standard in V.AI-10.4; added the corresponding Appendix C.1 and C.2 template fields. Reframed the AI-7.4 rationale to name the two threat cases (tampering after validation; hostile origin surviving integrity verification). Added AI-7.7 Artifact Load Safety, binding serialization formats and loader configurations on every artifact load path, with verification by inspection and negative test and a Section 4.2 matrix row. Rewrote the Section 1.2 scope statement and aligned Section 1.1 so that classification selects requirements rather than gating entry, consistent with Table 1-1. Adopted the version numbering

Version	Date	Author	Change Class	Description
				and change management policy stated above, added the Change Class column, and assigned the document identifier SCL-ARF-001. Removed "Conditionally Certified" from Appendix D.5 and stated the two-outcome rule, with Not Certified outcomes reporting per dimension and aggregated scores in the Technical Assessment Report and the Certification Determination Document. Corrected the Appendix A term count, added the Certified Configuration and Conformity Basis glossary entries, and settled the spelling of de-identification throughout.
4.0	2026-10 (public comment draft; deposit follows the comment window)	Safety Critical Labs	Normative Modification	Built from v3.7 under the 4.0 design brief of 2026-10-18. Sub-requirement identifiers in this description are those of version 3.7 (in versions 3.x numbering); the identifier map at the end of this section maps them to this version, in which AI-1, AI-2, AI-5 and AI-7 are renumbered to close the gaps the withdrawals left. Conformity model: evidence classes (deterministic, statistical, bounded, surveillance) made normative in Section 4.1 and named on every verification; insufficient evidence added as a verification outcome with its effect by tier stated in Section 1.6; the verification statement template (method, class, confirmations, discriminating criterion, declared population, artifact, cadence)

Version	Date	Author	Change Class	Description
				applied to every retained sub-requirement, with inspection no longer accepted for computed properties; a sufficiency statement required per requirement set (Appendix C.7); the residual list replaces the risk score on the certificate, and Appendix D becomes the evidence-strength profile reported in the Technical Assessment Report. Safety-Critical tier: new set AI-14 Runtime Assurance, mandatory at Safety-Critical, with monitor specification, assured fallback, switching timing, coverage statement and validation evidence; AI-4.4, AI-5.3, AI-6.3, AI-9.4 and AI-2.11 re-pointed to it; AI-9.1's autonomous-operation case made a hazard-analysis condition; AI-5.5 no longer accepts human review as the independent source at Safety-Critical; AI-13.1 confines Safety-Critical learning to shadow mode. Thresholds: Section 1.5 now requires a derivation basis for every value and states the adequacy criteria in the public text; Section 1.7 (new) requires declared evaluation populations with required strata; Section 5.1.1 values labelled illustrative and unsourced. Structure: Section 4.6 (new) states the monitoring, alerting, response and logging cycle once, withdrawing AI-2.6, AI-2.7, AI-2.9, AI-5.4, AI-6.4 and AI-7.5; AI-1.5 to AI-1.8 withdrawn and carried to new set AI-15; AI-2 rescoped to disparity

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				across declared strata with AI-2.8 and AI-2.10 rewritten; AI-5 made conditional on generative or free-form output; AI-3.1 defines coverage over the ODD grid and excludes structural network coverage; AI-3.2 requires a statistical power statement; AI-4.1 requires clean-stream qualification and in-service deployability; AI-4.2 made conditional on ground-truth availability; AI-7.2 rewritten around a declared threat model and budget with adaptive attacks at Safety-Critical; AI-8.2 and AI-12.3 require faithfulness; AI-9.7 to AI-9.9 conditional on the boundary including the operating organization. Classification authorities defined functionally and the four classification questions made normative (Section 1.2.2, Appendix B). New Section 1.0 Development and Governance. Glossary gains fourteen terms. Identifiers are stable within a major version and the identifier map below carries the mapping; the identifier disposition table is the first section of the 4.0 change log. The v3.7.1 patch items are folded in rather than released separately: the structural nouns (requirement set, sub-requirement, requirement area) defined in Appendix A, and the NPR 8000.4C lineage of the Appendix D dimensions cited in D.1, at R.AI-1.9 and in Table 1-10. Pre-window changes of 2026-10-23,

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				from a critique of AI-5 taken before the comment draft was announced: a new sub-requirement, AI-5.5 Hallucination Audit in 4.0 numbering, measures the audited miss rate in service on a sample of outputs that passed detection and validation (surveillance evidence); Section 4.6.1 requires detector recall on a declared seeded set run in shadow mode wherever an indicator counts detections; and AI-5.5 Independent Output Validation (AI-5.4 in 4.0) adds the measured joint miss rate where the independent source is itself a learned artifact and bars runtime validation sources outside change control unless validated under AI-1.10 (AI-1.6 in 4.0).